

Module 4: SNMP, ASN.1, Physical Layer: Data and signals, Digital transmission, Analog transmission, Bandwidth utilization, Transmission media

Network Management

- Network management is implemented at the **Application Layer** of the TCP/IP protocol suite.
- It ensures efficient operation, monitoring, and maintenance of computer networks.
- As the Internet grows larger, network management becomes increasingly important.
- Proper management improves:
 - Reliability
 - Availability
 - Performance
 - Security

Why Network Management is Important

- Modern networks consist of:
 - Routers
 - Switches
 - Servers
 - Wireless Devices
 - Hosts
- Failure of a single device may interrupt communication.
- Network administrators must continuously monitor the network.
- Early fault detection minimizes downtime.



Need for Network Management

- Internet size is continuously increasing.
- Large-scale networks require centralized management.
- Network management helps to:
 - Detect faults
 - Improve performance
 - Configure devices
 - Monitor traffic
 - Maintain security

Key Idea

Without proper network management, communication between different parts of the Internet may be interrupted.

Objectives of Network Management

- Ensure continuous network availability
- Detect and isolate network failures
- Improve network performance
- Configure and manage network devices
- Protect network resources from unauthorized access
- Maintain reliable communication

Topics Covered in this Chapter

This chapter is divided into three major sections:

- ① Areas of Network Management
- ② Network Management using SNMP
- ③ Structure of Management Information (SMI) and MIB

Learning Outcome

Understand how network management is implemented at the Application Layer of the TCP/IP protocol suite.

Network Management

- Network management involves:
 - Monitoring
 - Testing
 - Configuring
 - Troubleshooting
- Ensures smooth and efficient network operation
- Maintains the required Quality of Service (QoS)
- Uses:
 - Hardware
 - Software
 - Human administrators

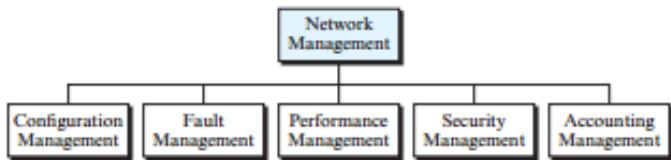
ISO Areas of Network Management

The ISO defines five major areas of network management.

FCAPS

F	Fault Management
C	Configuration Management
A	Accounting Management
P	Performance Management
S	Security Management

Figure 9.1 *Areas of network management*



1. Configuration Management

- Maintains network configuration information
- Handles changes in:
 - Hardware
 - Software
 - User Accounts
- Includes:
 - Reconfiguration
 - Documentation
- Maintains hardware maps, software versions, and user privileges



2. Fault Management

- Ensures proper network operation
- Two approaches:
 - Reactive Fault Management
 - Proactive Fault Management
- Reactive Fault Management:
 - 1 Detect Fault
 - 2 Isolate Fault
 - 3 Correct Fault
 - 4 Record Fault
- Proactive Fault Management predicts and prevents failures

3. Performance Management

- Monitors network efficiency
- Important performance metrics:
 - Capacity
 - Traffic
 - Throughput
 - Response Time
- Helps prevent network congestion
- Ensures optimal network performance



4. Security and Accounting Management

Security Management

- Controls access to network resources
- Uses:
 - Authentication
 - Encryption

Accounting Management

- Monitors resource usage
- Charges users or departments
- Prevents misuse of network resources
- Helps in future network planning

Summary: FCAPS Model

Area	Purpose
Configuration	Configure and document network devices
Fault	Detect, isolate and repair faults
Performance	Monitor capacity, traffic and throughput
Security	Protect network resources
Accounting	Track resource usage and billing

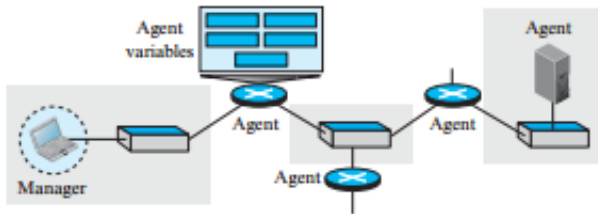
Exam Point

Remember the acronym **FCAPS** (Fault, Configuration, Accounting, Performance, Security).

Simple Network Management Protocol (SNMP)

- SNMP is the Internet standard for network management.
- Application layer protocol of the TCP/IP suite.
- Provides monitoring and maintenance of network devices.
- Uses Manager–Agent architecture.
- Supports heterogeneous LANs, WANs and devices.

Figure 9.2 *SNMP concept*



Managers and Agents

Manager

- Runs SNMP client.
- Monitors and controls agents.
- Retrieves and modifies management information.

Agent

- Runs SNMP server.
- Maintains management database.
- Sends Trap messages on abnormal events.

Three Basic Functions

- 1 Manager retrieves information.
- 2 Manager changes object values.
- 3 Agent reports abnormal events (Trap).

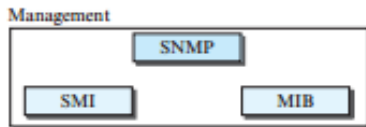
Management Components

Internet network management uses three components:

- SNMP
- SMI
- MIB



Figure 9.3 *Components of network management on the Internet*



Roles of SNMP, SMI and MIB

SNMP	Exchanges management messages and reads/writes object values.
SMI	Defines naming rules, data types and encoding.
MIB	Defines managed objects and their relationships.

Figure 9.4 *Comparing computer programming and network management*

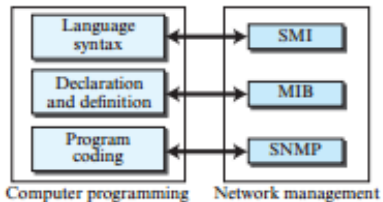
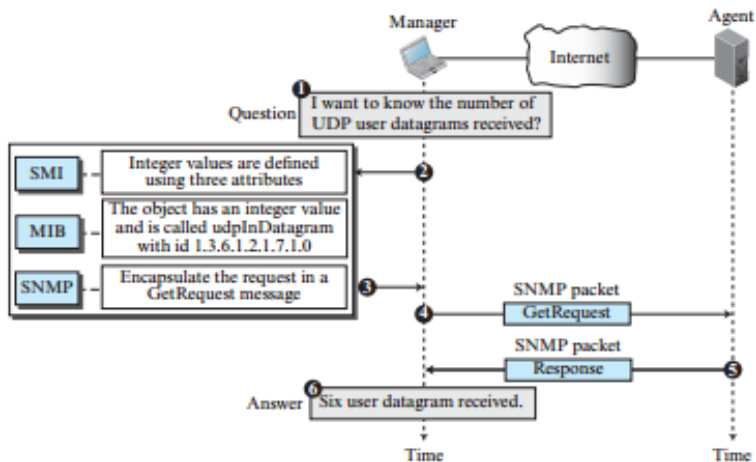


Figure 9.5 Management overview

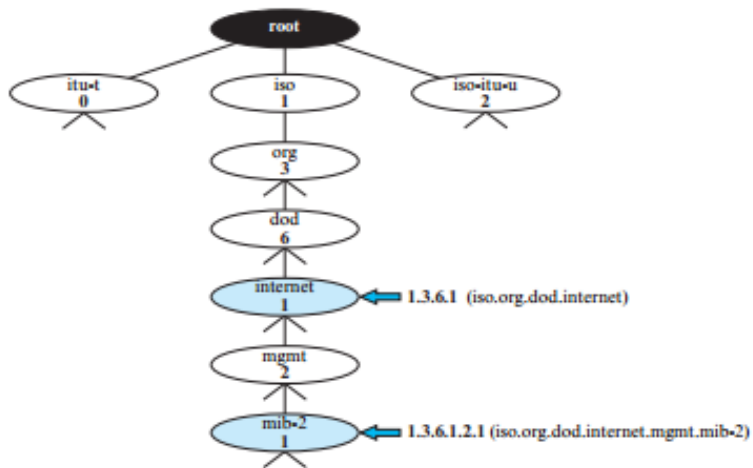


SMI defines:

- Object naming
- Data types
- Encoding rules



Figure 9.6 Object identifier in SMI



Object Identifier (OID)

- Hierarchical tree structure.
- Human-readable:
 - iso.org.dod.internet.mgmt.mib-2
- Numeric:
 - 1.3.6.1.2.1
- All MIB-II objects begin with 1.3.6.1.2.1.

SMI Data Types

Simple Types:

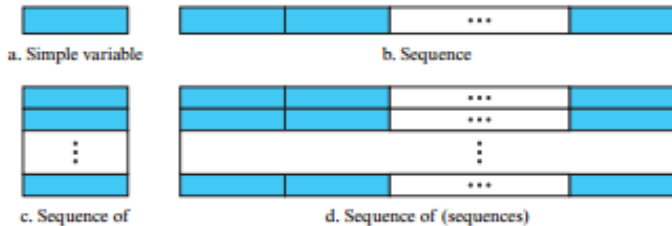
- INTEGER
- Integer32
- Unsigned32
- OCTET STRING
- OBJECT IDENTIFIER
- IPAddress
- Counter32
- Counter64
- Gauge32
- TimeTicks
- BITS
- Opaque

Structured Types:

- SEQUENCE
- SEQUENCE OF



Figure 9.7 Conceptual data types

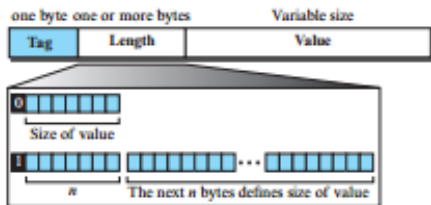


BER uses TLV format:

- Tag
- Length
- Value



Figure 9.8 *Encoding format*



- Database of managed objects.
- Objects follow SMI rules.
- Stores names and types.

Example 9.1

Figure 9.9 shows how to define INTEGER 14. The size of the length field is from Table 9.1.

Figure 9.9 Example 9.1: INTEGER 14

0x02	0x04	0x00	0x00	0x00	0x0E
Tag (integer)	Length (4 bytes)	Value (14)			

Example 9.2

Figure 9.10 shows how to define the OCTET STRING "HL".

Figure 9.10 Example 9.2: OCTET STRING "HL"

0x04	0x02	0x48	0x49
Tag (String)	Length (2 bytes)	Value (H)	Value (l)

Example 9.3

Figure 9.11 shows how to define ObjectIdentifier 1.3.6.1 (iso.org.dod.internet).

Figure 9.11 Example 9.3: ObjectIdentifier 1.3.6.1

0x06	0x04	0x01	0x03	0x06	0x01
Tag (ObjectId)	Length (4 bytes)	Value (1)	Value (3)	Value (6)	Value (1)
-----1.3.6.1 (iso.org.dod.internet)----->					

Example 9.4

Figure 9.12 shows how to define IPAddress 131.21.14.8.

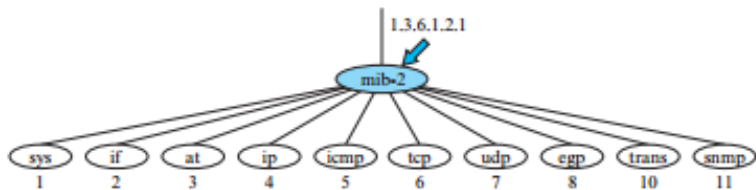
Figure 9.12 Example 9.4: IPAddress 131.21.14.8

0x40	0x04	0x83	0x15	0x0E	0x08
Tag (IPAddress)	Length (4 bytes)	Value (131)	Value (21)	Value (14)	Value (8)
-----131.21.14.8----->					

MIB-II Groups

sys, if, at, ip, icmp, tcp, udp, egp, transmission, snmp

Figure 9.13 *Some mib-2 groups*



Accessing MIB Variables

Example:

- udpInDatagrams
- OID: 1.3.6.1.2.1.7.1.0

Figure 9.14 *udp group*

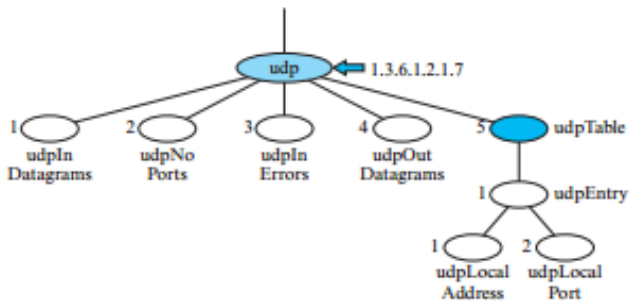


Figure 9.15 *udp variables and tables*

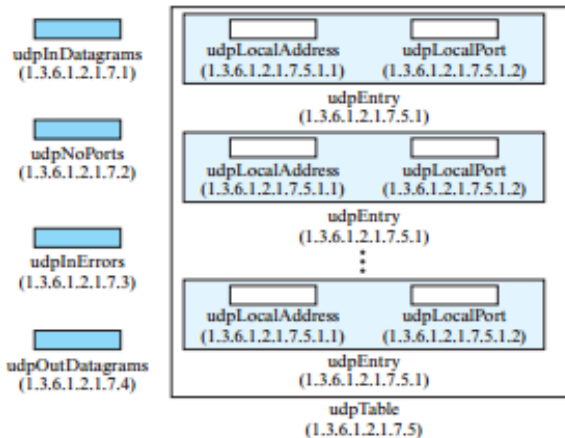


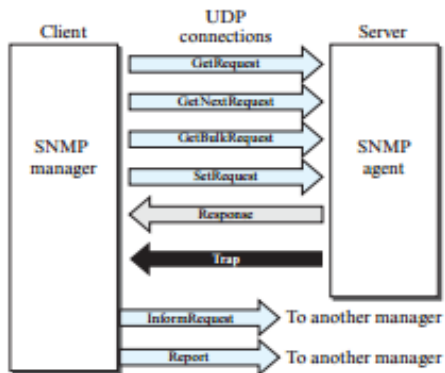
Figure 9.16 Indexes for udpTable

181.23.45.14	23
1.3.6.1.2.1.7.5.1.1.181.23.45.14.23	1.3.6.1.2.1.7.5.1.2.181.23.45.14.23
192.13.5.10	161
1.3.6.1.2.1.7.5.1.1.192.13.5.10.161	1.3.6.1.2.1.7.5.1.2.192.13.5.10.161
227.2.45.18	180
1.3.6.1.2.1.7.5.1.1.227.2.45.18.180	1.3.6.1.2.1.7.5.1.2.227.2.45.18.180
230.20.5.24	212
1.3.6.1.2.1.7.5.1.1.230.20.5.24.212	1.3.6.1.2.1.7.5.1.2.230.20.5.24.212

- GetRequest
- GetNextRequest
- GetBulkRequest
- SetRequest
- Response
- Trap
- InformRequest
- Report



Figure 9.17 *SNMP PDUs*

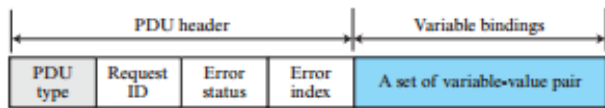


Fields:

- PDU Type
- Request ID
- Error Status
- Error Index
- Variable Bindings



Figure 9.18 SNMP PDU format



a. All PDU types except GetBulkRequest



b. GetBulkRequest

Note:

The error status and error index values are set to 0 for all request messages.

Figure 9.19 *SNMP message*

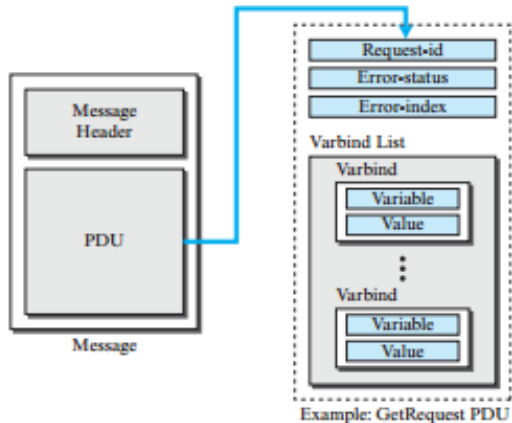


Figure 9.20 Example 9.5

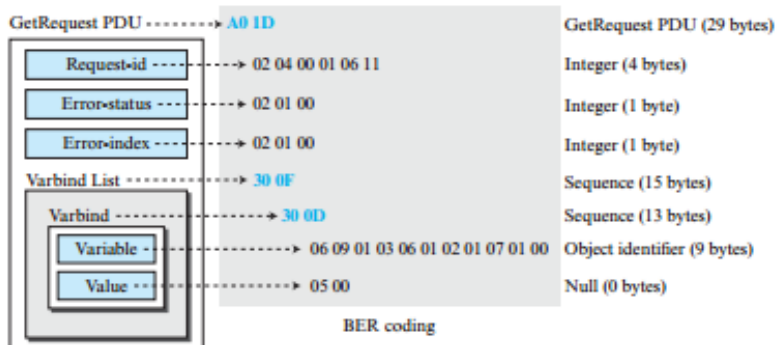


Figure 9.21 *Actual message sent for Example 9.5*

30	29	**	**
**	**	**	**
**	**	**	**
A0	1D	02	04
00	01	06	11
02	01	00	02
01	00	30	0F
30	0D	06	09
01	03	06	01
02	01	07	01
00	05	00	

Message

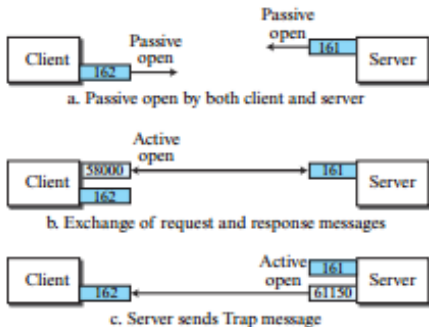
Note:

The byte values
are in hexadecimal

UDP Ports

- Agent: Port 161
- Manager receives Trap: Port 162
- Request/Response uses UDP.

Figure 9.22 Port numbers for SNMP



Features:

- Authentication
- Confidentiality
- Integrity
- Remote security administration



Summary

- SNMP manages network devices.
- SMI defines naming and encoding.
- MIB stores managed objects.
- Manager-Agent communication uses UDP.
- SNMPv3 improves security.

- Explain ASN.1 and its purpose.
- Describe ASN.1 naming and syntax.
- Understand simple and structured data types.
- Explain BER encoding.
- Relate ASN.1 with SNMP, SMI and MIB.

What is ASN.1?

- ASN.1 stands for Abstract Syntax Notation One.
- International standard language for describing data.
- Platform independent.
- Used by SNMP, LDAP, X.509, PKI and telecommunications.
- Defines data structure, not bit representation.

Need for ASN.1

- Common syntax for heterogeneous systems.
- Eliminates ambiguity during communication.
- Supports interoperability.
- Independent of programming language and hardware.

- Object definitions
- Data types
- Constraints
- Encoding rules
- Used together with BER, DER, CER and PER.



Common Keywords

BEGIN

INTEGER

OCTET STRING

SEQUENCE

CHOICE

END

BOOLEAN

OBJECT IDENTIFIER

SET

NULL

Simple Data Types

- INTEGER
- BOOLEAN
- OCTET STRING
- OBJECT IDENTIFIER
- NULL
- REAL

Structured Data Types

- SEQUENCE
- SEQUENCE OF
- SET
- SET OF
- CHOICE



SEQUENCE

- Similar to C structure.
- Collection of fields.
- Fields may have different types.

SEQUENCE OF

- Similar to an array.
- All elements have same type.

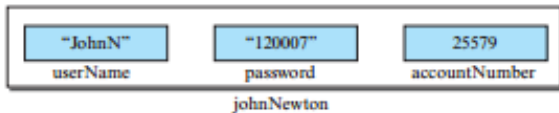
CHOICE

- Only one alternative is selected.
- Similar to union in C.

Object Identifier (OID)

- Hierarchical naming.
- Example:
 - iso.org.dod.internet.mgmt.mib-2
- Numeric form: 1.3.6.1.2.1

Figure 9.23 *Record representing the type definition and variable declaration*



Encoding Rules

- BER - Basic Encoding Rules
- DER - Distinguished Encoding Rules
- CER - Canonical Encoding Rules
- PER - Packed Encoding Rules

BER encodes data using TLV:

- Tag
- Length
- Value



- SMI uses ASN.1.
- Defines object names.
- Defines data types.
- Defines encoding.
- Supports MIB objects.



Advantages

- Vendor independent.
- Platform independent.
- Standardized.
- Easy interoperability.
- Supports complex data.



Summary

- ASN.1 is a universal data description language.
- Used extensively in SNMP.
- Supports simple and structured data.
- BER converts ASN.1 objects into binary.
- Foundation for network management.